

Degree: - B.Tech. Computer Science and Engineering (Big data Analytics)
Semester:6
END-SEMESTER EXAMINATION, APRIL-MAY 2026

Course Code: CBCPE601**Course Title: Cloud Computing****Time: 3HRS****Max Marks:50**

Note: - Attempt all the five Questions. Missing data/ information if any, maybe suitably assumed and mentioned in the answer.

Q.No	Questions	Marks	CO
1.	Attempt any 2 parts of the following		
1a.	Discuss the key characteristics of cloud computing and explain how they improve scalability and cost optimisation compared to traditional systems.	5	CO1
1b.	Compare IAAS, PAAS, and SAAS in detail with architecture diagrams and real-world use cases.	5	CO1
1c.	Design a cloud adoption strategy for a mid-sized enterprise, including deployment models, benefits, and potential limitations.	5	CO1
2.	Attempt any 2 parts of the following		
2a.	Compare SOAP and REST web services in terms of performance, scalability, and security in cloud computing Environment.	5	CO2
2b.	Explain the working of hypervisors Type 1 and Type 2 and explain their role in virtualization.	5	CO2
2c.	Analyse how AJAX and Web Mashups contribute to dynamic cloud-based applications with suitable examples.	5	CO2
3.	Attempt any 2 parts of the following		
3a.	Explain virtual machine architecture and analyse its role in resource sharing and isolation.	5	CO3
3b.	Compare multi-tenancy and multi-entity support models with respect to scalability and security.	5	CO3
3c.	Discuss the pitfalls of virtualization and propose practical solutions for enterprise systems.	5	CO3
4.	Attempt any 2 parts of the following		
4a.	Explain the Map-Reduce programming model with a suitable example and analyse its efficiency in big data processing.	5	CO4
4b.	Compare GFS, HDFS, HBase, BigTable, and Dynamo in terms of architecture, and use cases.	5	CO4
4c.	Compare Mobile cloud computing, Sky computing, Utility Computing and elastic computing.	5	CO4
5.	Attempt any 2 parts of the following		
5a.	Discuss major security and privacy challenges in cloud computing and provide a secure architecture.	5	CO5
5b.	Explain Quality of Service in cloud computing and discuss methods for monitoring and maintaining Quality of Service.	5	CO5
5c.	Explain load balancing and resource optimization techniques for efficient cloud performance.	5	CO5